

EDUCATION

Amrita Vishwa Vidyapeetham

Coimbatore, TN

Bachelor of Technology in Computer Science and Engineering; CGPA: 7.78

Aug 2023 – Aug 2027

- **Core Coursework:** Data Structures & Algorithms (DSA), Operating Systems, DBMS, Computer Networks, Theory of Computation, Design and Analysis of Algorithms (DAA).
- **Specialized Coursework:** Machine Learning, Full Stack Frameworks, Design Patterns, Embedded Systems, Computer Organization & Architecture, User Interface Design.

TECHNICAL SKILLS

Languages: Python, C, Java, JavaScript, HTML, CSS

Frameworks & Paradigms: Node.js, Express.js, Full Stack Development, Design Patterns, OOP

Hardware & Embedded: STM32 Microcontrollers, Digital Electronics, Embedded Systems

Tools: Git, GitHub, VS Code, Postman

Interests: Machine Learning, Web Development, Competitive Programming, UI/UX Design

PROJECTS

Emergency Response Dispatch System | *Data Structures, Graphs, C++/Java*

- Engineered a city-scale simulation for emergency services (Police, Ambulance, Fire) using **Graph Theory** with city locations as nodes and routes as weighted edges.
- Implemented **Dijkstra's Algorithm** and **Priority Queues** to optimize responder dispatch times based on real-time emergency severity and distance.
- Utilized **Hash Maps** for $O(1)$ location lookups and status tracking, significantly improving simulated response efficiency.

Full-Stack Quiz Platform | *Node.js, Express, HTML/CSS, JavaScript*

- Developed a comprehensive web application featuring a dynamic frontend and a robust backend to host and manage interactive quizzes.
- Implemented user authentication and real-time score calculation logic, ensuring a seamless and secure testing environment.

PPT Controller using STM32 | *Embedded Systems, C, STM32 Microcontroller*

- Engineered a hardware-based interface to wirelessly or via serial communication control PowerPoint presentations using an STM32 board.
- Programmed microcontroller logic to map physical inputs to keyboard shortcuts, enhancing presentation accessibility and efficiency.

Personal Finance Tracker | *Python, Data Structures*

- Built a console-based tool to track expenses using custom data structures, optimizing for memory efficiency and persistent storage via file handling.

Thermostat Fire Detection System | *Embedded Systems, Sensors*

- Designed a temperature-sensitive detection system that triggers automated visual alerts when safety thresholds are reached.

Fitness Website | *Full Stack Frameworks, UI/UX Design*

- Developed a responsive web application providing personalized workout plans, utilizing UI design principles for enhanced user experience.

CERTIFICATIONS & ACADEMIC HONORS

Technical Workshops: Participated in Design and Analysis of Algorithms Workshop organized by IISC Bangalore.

Hardware Proficiency: Skilled in digital circuit simulation and troubleshooting via Digital Electronics Laboratory work.

Self-Paced Learning: Completed foundational courses in Machine Learning and Cyber-Security from UNSTOP.